



Cestodes

Taenia spp.

GIT module- Microbiology- LG3

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Objectives

- Two *Taenia* spp. Namely *Taenia saginata* and *Taenia solium* are of medical importance causing gastrointestinal infestation and the cysticerci of *T. solium* causes serious condition known as cysticercosis that can be fatal.
- To learn the morphology , life cycle, transmission, pathogenicity, clinical manifestation, diagnosis , and preventive measures.

List of Contents:

- Introduction
- Morphology
- Life cycle
- Pathogenicity and immunobiology
- Clinical manifestations
- Diagnosis
- Treatment
- Prevention and control

Introduction

- Approximately 45 *Taenia* spp. have been identified of those only two spp. namely *Taenia saginata* and *Taenia solium* are commonly responsible for human infection.
(*T. asiatica*)
- Common name: Pork tapeworm (*T. solium*), beef tapeworm (*T. saginata*)
- Taeniasis, Cysticercosis.
- 100 million people are infected worldwide, with major endemic areas located primarily in the developing countries of South America, Africa, India, china, and Southeast Asia.

- The total number of people suffering from NCC, including symptomatic and asymptomatic cases, is estimated to be between 2.56–8.30 million, based on the range of epilepsy prevalence data available. 70% of patients with epilepsy could lead a normal life if treated correctly.
- *T. solium* is the cause of 30% of epilepsy cases in many endemic areas where people and roaming pigs live in close proximity.

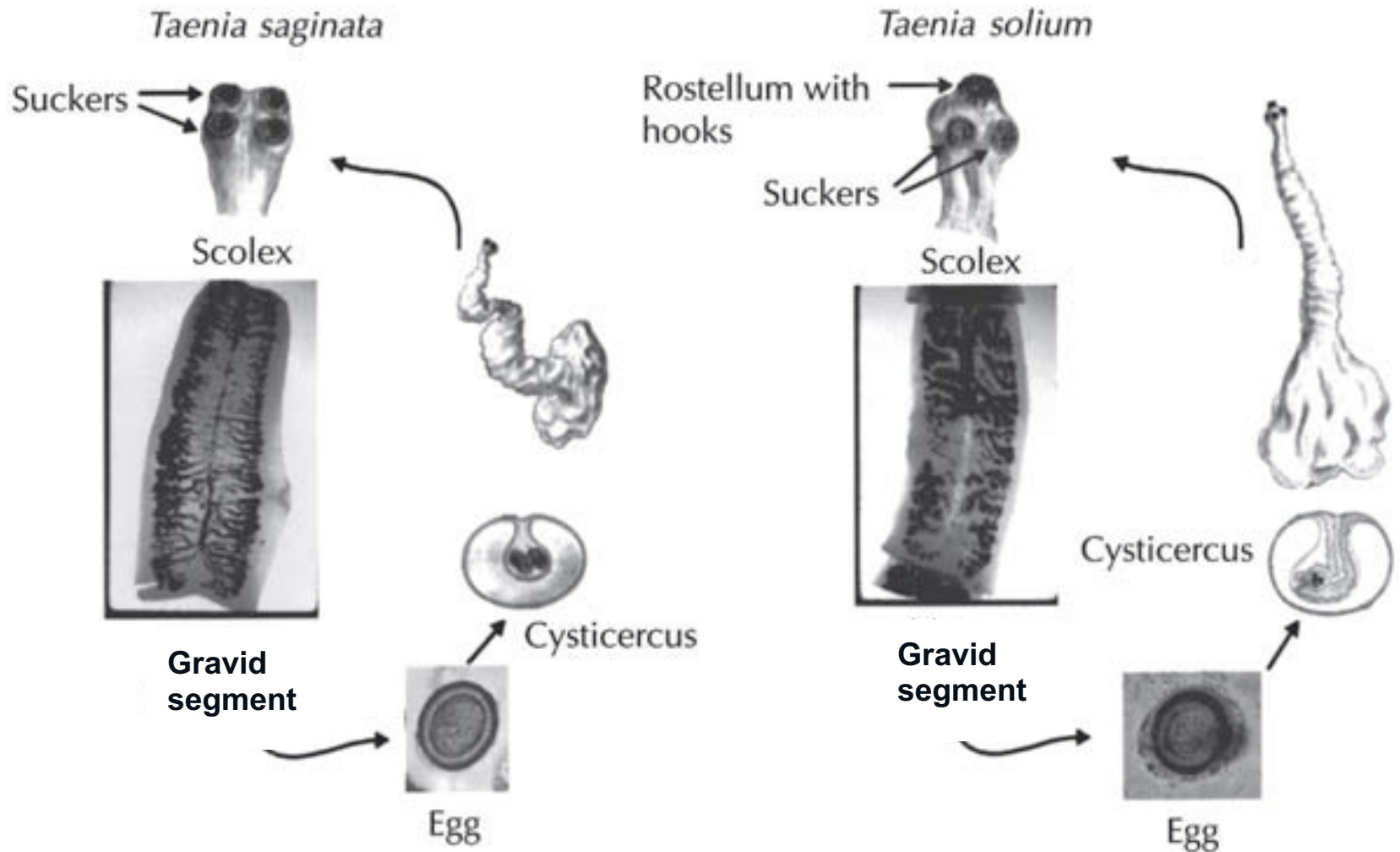
- Habitat
- Infective stage (*Cysticercus bovis*, *C. cellulosae*)
- Hosts:

Intermediate hosts: cow, pig

Definitive host: human

Morphology

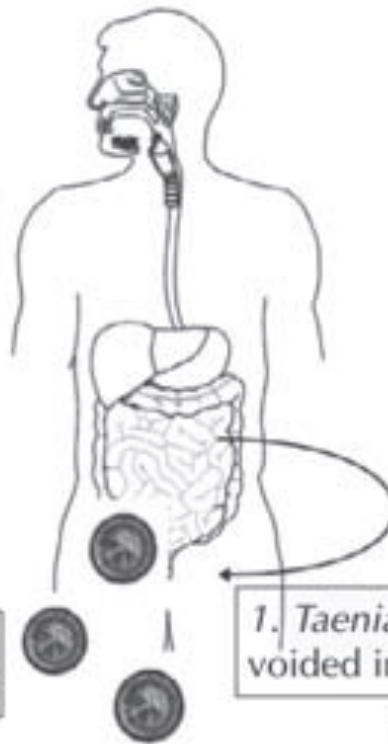
Comparative Morphology of *Taeniarhynchus saginatus* and *Taenia solium*



4. Ingested cysticerci reach intestine, attach and grow to adult tapeworms

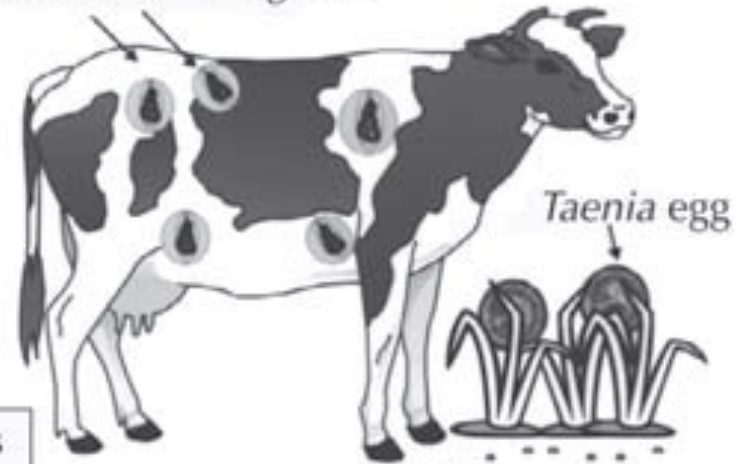


3. Infected meat inadequately cooked



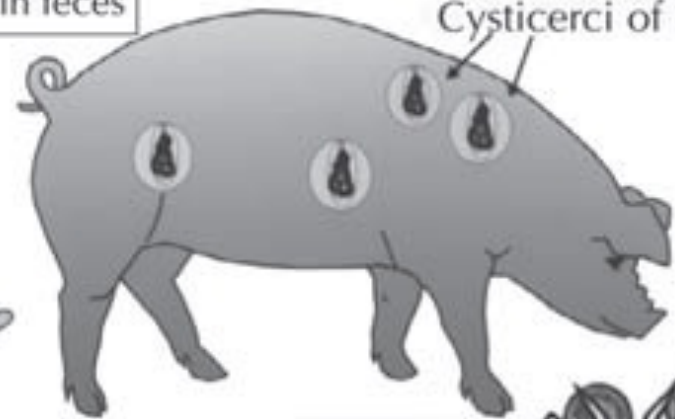
1. *Taenia* eggs voided in feces

Cysticerci of *T. saginata*

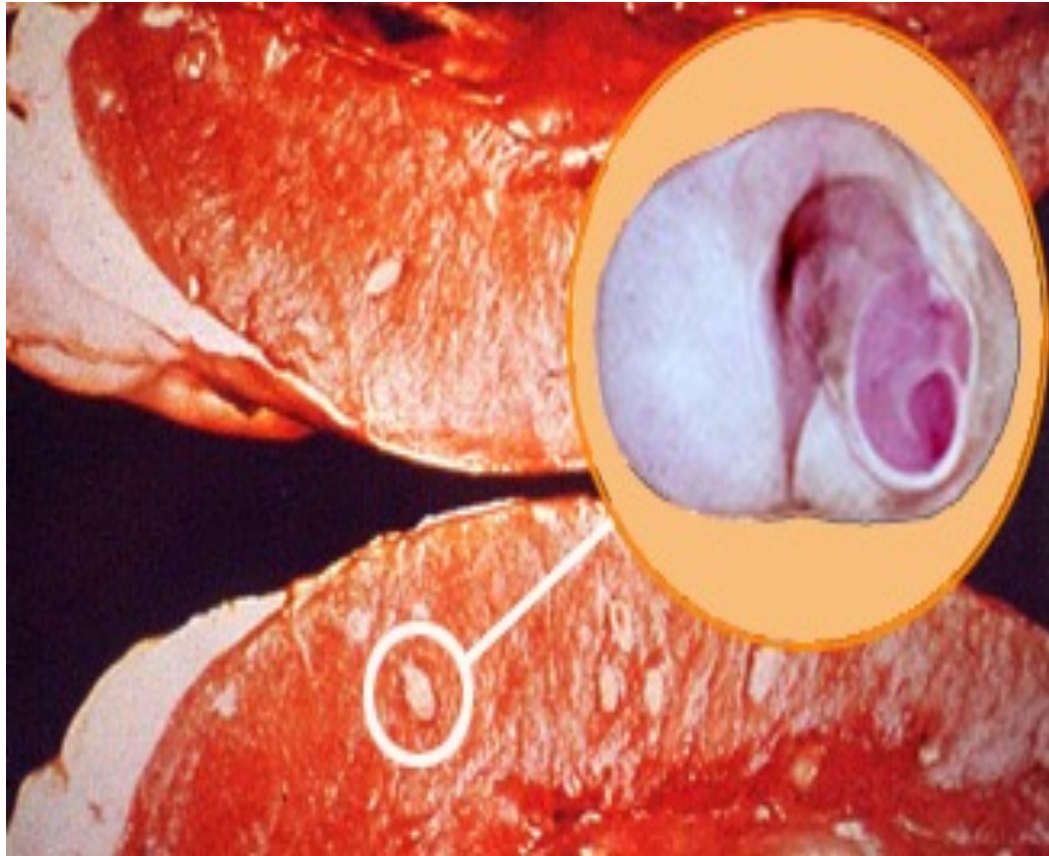


Taenia egg

Cysticerci of *T. solium*



2. Eggs ingested, cysticerci develop in tissues and organs of intermediate host



Cysticercus

Pathogenesis

- Adult worm may cause irritation or inflammation of the intestinal mucosa.
- In the case of *T. solium* the larval stage can develop in the muscle and some crucial organs such as brain and eye causing cysticercosis which can be fatal

Clinical manifestations

I- Taeniasis:

- Incubation periods (6-8 weeks),
- Is infection with adult worms and result in minor gastrointestinal irritation and is frequently accompanied by nausea, vomiting, diarrhea, constipation, hunger pain, chronic indigestion and **passage of the proglottids in the feces.**

II- Cysticercosis:

Refers to the tissue infection caused by the metacestode or larval stage of *T. solium* and is **acquired by accidentally ingestion of eggs or autoinfection**. Most often localize the skeletal muscles, eye and brain.

Clinical manifestations are direct result of inflammatory response to control the parasite growth, and dependent upon a variety of factors including the site of infection and the number of cysticerci within the tissue.

- Patients suffering from ocular cysticercosis frequently experience pain in the eyes accompanied by blurriness and partial or complete loss of vision.
- Patients infected with cysticerci in the skeletal muscles and or subcutaneous tissues are usually asymptomatic. Cysts range from 10-15 mm in length. Destruction of the cyst induce strong inflammatory response accompanied by localized pain and swelling

- Neurocysticercosis: is the most common parasite infection of central nervous system (60-90%) of infected individuals. May range from 4-20 mm in length. Destruction of the parasite induces strong inflammation, granuloma formation and fibrosis which may result in a sub acute encephalitis. Seizure is the most common symptom reported in patients with neurocysticercosis. other commonly associated clinical manifestations. include headache, dizziness, involuntary muscle movement, intracranial hypertension and dementia. Certain patients with cysticerci never develop any symptoms.

- **Extra-parenchymal/Subarachnoid Neurocysticercosis (cysts in spaces around the brain or ventricles):** This severe form carries a much higher mortality risk, with a reported death rate of roughly 16.6% to over 20% if complicated by severe inflammation, intracranial pressure, or stroke.
- **Parenchymal Neurocysticercosis (cysts in brain tissue):** Survival is very high, with a mortality rate under 2% for viable or calcified single/low-burden parenchymal cases. Patients with a single lesion have an excellent chance of full recovery.

Immunobiology of cysticerci

- Induce little or no inflammatory response
- Serine-threonine protease inhibitor, taeniastatine, inhibit complement activation, block cytokine production and interferes with neutrophil function.
- Para-myosin impairs complement system, primarily through inhibiting the activity of C1q.
- Activated complement is directed away from the parasite by the production of sulfated polysaccharide.

- Antibodies produced by the host bind the metacystode form through Fc receptor then are degraded and utilized as a source of amino acids for the parasite.
- Glutathione-S transferase and other small molecules produced by the cyst form are involved in the detoxification of reactive oxygen and nitrogen intermediates.

Diagnosis

- Demonstration of proglottids or free characteristic eggs in the fecal specimen.
- For cysticercosis, the definitive diagnosis is based on the combination of serological and epidemiological data.
- Imaging techniques: MRI and CT scan are considered to be the most sensitive method of detection of neurocysticercosis.

Treatment

- Praziquantel: at 5-10 mg/kg in a single oral dose.
- Niclosamide at an oral dose of 2 g for an adult given in a divided doses over 2 days.
- For cysticercosis: Praziquantel, Albendazole, flubendazole and anti-inflammatory and anticonvulsants.

Prevention and control

Personal protection is by thorough cooking beef and pork (above 56°C) or by freezing at -10°C for 10 days

References

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Thanks a lot